Theorema Egregium

All surfaces, maps and vector fields are smooth. Let $\Sigma \subseteq \mathbb{R}^3$ be a regular surface and let $X\colon \Omega \to U \subseteq \Sigma$ be a local parametrization, where $\Omega \subseteq \mathbb{R}^2$ is open. A tangent field along X is a map $Y\colon \Omega \to \mathbb{R}^3$ with $Y(p) \in T_{X(p)}\Sigma$. Put

$$N = \frac{X_u \times X_v}{\|X_u \times X_v\|}, \qquad E = \langle X_u, X_u \rangle, \quad F = \langle X_u, X_v \rangle, \quad G = \langle X_v, X_v \rangle,$$

and $e = \langle X_{uu}, N \rangle$, $f = \langle X_{uv}, N \rangle$, $g = \langle X_{vv}, N \rangle$. We write I and II for the first and second fundamental forms. Their matrices in the frame (X_u, X_v) are

$$[\mathrm{I}] = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \qquad [\mathrm{II}] = \begin{pmatrix} e & f \\ f & g \end{pmatrix}.$$

For a column $c = (a, b)^t$, one has $c^t[\mathrm{I}]c = \|aX_u + bX_v\|^2$. Regularity makes this quantity positive for $c \neq 0$. In particular, $D = \det[\mathrm{I}] = EG - F^2 > 0$. The map $N\colon \Omega \to \mathbb{S}^2$ defines the unit normal $\nu = N \circ X^{-1}$ on U. Since $\langle \nu, \nu \rangle = 1$, the image of $\mathrm{d}\nu_q$ lies in $\nu(q)^\perp = T_q\Sigma$. The shape operator is

$$S_q\colon T_q\Sigma \to T_q\Sigma, \qquad w \mapsto -\,\mathrm{d}\nu_q(w), \qquad q \in U.$$

We use the definition $K(q) = \det S_q$. In coordinate formulas, K means $K \circ X$, and $S(Y)$ means the field $p \mapsto S_{X(p)}(Y(p))$.

1. Covariant derivatives

Definition 1.1. For a tangent field Y, a point $p \in \Omega$ and a vector $\xi \in \mathbb{R}^2$, define

$$(\nabla_\xi Y)_p = (\mathrm{d}Y)_p(\xi) - N(p)\langle (\mathrm{d}Y)_p(\xi), N(p) \rangle.$$

We write $\nabla_u Y$ and $\nabla_v Y$ for the derivatives in the parameter directions $(1, 0)$ and $(0, 1)$.

For example,

$$\nabla_u X_u = X_{uu} - eN, \qquad \nabla_v X_v = X_{vv} - gN, \qquad \nabla_u X_v = \nabla_v X_u = X_{uv} - fN.$$

Lemma 1.2. *If $\alpha\colon \Omega \to \mathbb{R}$ and Y, Z are tangent fields, then*

$$\nabla(\alpha Y) = \alpha \nabla Y + Y\,\mathrm{d}\alpha, \qquad \mathrm{d}\langle Y, Z \rangle = \langle \nabla Y, Z \rangle + \langle Y, \nabla Z \rangle.$$

Proof. For $\xi \in \mathbb{R}^2$, the ordinary product rule and $\langle Y, N \rangle = 0$ give

$$\nabla_\xi(\alpha Y) = \alpha\big(\mathrm{d}Y(\xi) - \langle \mathrm{d}Y(\xi), N \rangle N\big) + (\mathrm{d}\alpha)(\xi)Y.$$

This proves the first identity. For the second, the ordinary product rule gives

$$\mathrm{d}\langle Y, Z \rangle(\xi) = \langle \mathrm{d}Y(\xi), Z \rangle + \langle Y, \mathrm{d}Z(\xi) \rangle.$$

The normal components of $\mathrm{d}Y(\xi)$ and $\mathrm{d}Z(\xi)$ are orthogonal to Z and Y, respectively. They may therefore be removed from the two inner products. □

Write $X_1 = X_u$, $X_2 = X_v$, $u^1 = u$, $u^2 = v$, $\partial_i = \partial/\partial u^i$, and $X_{ij} = \partial_i \partial_j X$. The Christoffel symbols are defined by

$$X_{ij} = \Gamma^1_{ij} X_u + \Gamma^2_{ij} X_v + [\mathrm{II}]_{ij} N.$$

Set $\Gamma_i = (\Gamma_{ij}^k)_{k,j=1,2}$, with the upper index labelling the row. Thus, for the row frame $[X_u, X_v]$,

$$\nabla_u[X_u, X_v] = [X_u, X_v]\Gamma_1, \qquad \nabla_v[X_u, X_v] = [X_u, X_v]\Gamma_2.$$

Every tangent field can be written uniquely as $Y = aX_u + bX_v = [X_u, X_v]y$, where $y = (a, b)^t$. To recover the coefficients, take the inner products with X_u and X_v. They satisfy

$$[\mathrm{I}]y = \begin{pmatrix} \langle Y, X_u\rangle \\ \langle Y, X_v\rangle \end{pmatrix}, \qquad [\mathrm{I}]^{-1} = \frac{1}{D}\begin{pmatrix} G & -F \\ -F & E \end{pmatrix}.$$

The coefficients a, b are smooth. Applying the product rule column by column gives

$$\nabla_u Y = (\nabla_u[X_u, X_v])y + [X_u, X_v]\partial_u y = [X_u, X_v](\Gamma_1 y + \partial_u y).$$

Similarly, $\nabla_v Y = [X_u, X_v](\Gamma_2 y + \partial_v y)$. Thus

$$\nabla_u Y = [X_u, X_v](\partial_u + \Gamma_1)y, \qquad \nabla_v Y = [X_u, X_v](\partial_v + \Gamma_2)y. \tag{1.1}$$

Proposition 1.3. *The matrices Γ_1, Γ_2 are determined by E, F, G and their first derivatives:*

$$\Gamma_1 = \tfrac{1}{2}[\mathrm{I}]^{-1}\begin{pmatrix} E_u & E_v \\ 2F_u - E_v & G_u \end{pmatrix}, \qquad \Gamma_2 = \tfrac{1}{2}[\mathrm{I}]^{-1}\begin{pmatrix} E_v & 2F_v - G_u \\ G_u & G_v \end{pmatrix}.$$

Proof. For each i, j, k, taking the inner product of the decomposition of X_{ij} with X_k yields $\langle X_{ij}, X_k\rangle = \sum_\ell [\mathrm{I}]_{k\ell}\Gamma_{ij}^\ell$. The product rule and the equality of mixed derivatives give

$$\begin{aligned} \partial_i[\mathrm{I}]_{jk} &= \langle X_{ij}, X_k\rangle + \langle X_j, X_{ik}\rangle, \\ \partial_j[\mathrm{I}]_{ik} &= \langle X_{ij}, X_k\rangle + \langle X_i, X_{jk}\rangle, \\ \partial_k[\mathrm{I}]_{ij} &= \langle X_{ik}, X_j\rangle + \langle X_i, X_{jk}\rangle. \end{aligned}$$

Adding the first two lines and subtracting the third leaves $2\langle X_{ij}, X_k\rangle$. The six scalar products needed below are

$$\begin{aligned} \langle X_{uu}, X_u\rangle &= \tfrac{1}{2}E_u, & \langle X_{uu}, X_v\rangle &= F_u - \tfrac{1}{2}E_v, \\ \langle X_{uv}, X_u\rangle &= \tfrac{1}{2}E_v, & \langle X_{uv}, X_v\rangle &= \tfrac{1}{2}G_u, \\ \langle X_{vv}, X_u\rangle &= F_v - \tfrac{1}{2}G_u, & \langle X_{vv}, X_v\rangle &= \tfrac{1}{2}G_v. \end{aligned}$$

The columns of $[\mathrm{I}]\Gamma_i$ consist of these scalar products. Hence

$$[\mathrm{I}]\Gamma_1 = \frac{1}{2}\begin{pmatrix} E_u & E_v \\ 2F_u - E_v & G_u \end{pmatrix}, \qquad [\mathrm{I}]\Gamma_2 = \frac{1}{2}\begin{pmatrix} E_v & 2F_v - G_u \\ G_u & G_v \end{pmatrix}.$$

Multiplication by $[\mathrm{I}]^{-1}$ proves the formulas. □

2. Curvature and the Gauss equation

Definition 2.1. For a tangent field Y, define

$$RY = \nabla_u\nabla_v Y - \nabla_v\nabla_u Y.$$

Proposition 2.2. *Put $A = \partial_u\Gamma_2 - \partial_v\Gamma_1 + [\Gamma_1, \Gamma_2]$, where $[\Gamma_1, \Gamma_2] = \Gamma_1\Gamma_2 - \Gamma_2\Gamma_1$. If $Y = [X_u, X_v]y$, then*

$$RY = [X_u, X_v]Ay.$$

For each $p \in \Omega$, there is a unique linear map

$$R_p\colon T_{X(p)}\Sigma \to T_{X(p)}\Sigma$$

such that $R_p(Y(p)) = (RY)(p)$ for every tangent field Y along X.

Proof. By (1.1), the coordinate column of $\nabla_u \nabla_v Y$ is

$$\partial_u \partial_v y + (\partial_u \Gamma_2) y + \Gamma_2 \partial_u y + \Gamma_1 \partial_v y + \Gamma_1 \Gamma_2 y.$$

In the reverse order the coordinate column is

$$\partial_v \partial_u y + (\partial_v \Gamma_1) y + \Gamma_1 \partial_v y + \Gamma_2 \partial_u y + \Gamma_2 \Gamma_1 y.$$

The mixed derivatives of y agree. The terms containing $\partial_u y$ and $\partial_v y$ also agree in the two columns. Subtracting leaves the derivatives of the matrices and their two ordered products. It follows that

$$RY = [X_u, X_v]\big(\partial_u \Gamma_2 - \partial_v \Gamma_1 + \Gamma_1 \Gamma_2 - \Gamma_2 \Gamma_1\big) y = [X_u, X_v] A y.$$

Fix $p \in \Omega$ and $w \in T_{X(p)}\Sigma$. Write $w = aX_u(p) + bX_v(p)$. The tangent field $Y = aX_u + bX_v$ satisfies $Y(p) = w$. If $Z = [X_u, X_v]z$ is any other tangent field with $Z(p) = w$, uniqueness of coordinates gives $z(p) = (a, b)^t = y(p)$. The formula above therefore gives $(RZ)(p) = (RY)(p)$.

We may consequently define

$$R_p \colon T_{X(p)}\Sigma \to T_{X(p)}\Sigma, \qquad w \mapsto (RY)(p),$$

where Y is any tangent field along X satisfying $Y(p) = w$. In coordinates, $R_p(w) = [X_u(p), X_v(p)] A(p)(a, b)^t$. This formula proves linearity. It also proves uniqueness, since every w is the value of such a field. □

For a tangent field Y, write its ordinary derivatives as

$$Y_u = \nabla_u Y + h_u(Y) N, \qquad Y_v = \nabla_v Y + h_v(Y) N, \quad h_u(Y) = \langle Y_u, N \rangle, \quad h_v(Y) = \langle Y_v, N \rangle.$$

Differentiating $\langle Y, N \rangle = 0$ gives $h_u(Y) = -\langle Y, N_u \rangle$ and $h_v(Y) = -\langle Y, N_v \rangle$. Since $\langle N, N \rangle = 1$, both N_u and N_v are tangent. The chain rule gives $\mathrm{d}N_p = \mathrm{d}\nu_{X(p)} \circ \mathrm{d}X_p$. Hence the shape operator at $X(p)$ is the linear map

$$S_{X(p)} \colon T_{X(p)}\Sigma \to T_{X(p)}\Sigma, \qquad aX_u(p) + bX_v(p) \mapsto -aN_u(p) - bN_v(p).$$

We write $S(X_u) = -N_u$ and $S(X_v) = -N_v$. Then

$$h_u(Y) = \langle S(X_u), Y \rangle, \qquad h_v(Y) = \langle S(X_v), Y \rangle.$$

Differentiating $\langle X_j, N \rangle = 0$ gives $\langle S(X_i), X_j \rangle = \langle X_{ij}, N \rangle = [\mathrm{II}]_{ij}$. Thus $h_u(X_u) = e$, $h_u(X_v) = h_v(X_u) = f$, and $h_v(X_v) = g$. In particular, $\langle S(X_j), X_i \rangle = [\mathrm{II}]_{ij}$ for $i, j \in \{1, 2\}$. Since $[\mathrm{II}]$ is symmetric and X_u, X_v span the tangent plane, S is self-adjoint. If B is the matrix of S in this frame, its jth column gives the coordinates of $S(X_j)$. Pairing with each X_i gives $[\mathrm{I}]B = [\mathrm{II}]$. Thus $B = [\mathrm{I}]^{-1}[\mathrm{II}]$ and

$$K = \det B = \frac{\det[\mathrm{II}]}{\det[\mathrm{I}]} = \frac{eg - f^2}{D}.$$

Proposition 2.3 (Gauss equation)**.** *For every tangent field Y,*

$$RY = h_v(Y) S(X_u) - h_u(Y) S(X_v). \tag{2.1}$$

Proof. Decompose $Y_i = \nabla_i Y + h_i(Y)N$, for $i = u, v$. For $i, j \in \{u, v\}$, differentiating the expression for Y_j in the i direction gives

$$\partial_i Y_j = \partial_i(\nabla_j Y) + (\partial_i h_j(Y))N + h_j(Y)N_i.$$

The field $\nabla_j Y$ is tangent, so its derivative decomposes as

$$\partial_i(\nabla_j Y) = \nabla_i \nabla_j Y + h_i(\nabla_j Y)N.$$

Substituting this identity and $N_i = -S(X_i)$ gives

$$\partial_i \partial_j Y = \nabla_i \nabla_j Y - h_j(Y)S(X_i) + \big(h_i(\nabla_j Y) + \partial_i h_j(Y)\big)N.$$

The tangent components in the two orders are therefore

$$(Y_{uv})^\top = \nabla_u \nabla_v Y - h_v(Y)S(X_u), \qquad (Y_{vu})^\top = \nabla_v \nabla_u Y - h_u(Y)S(X_v).$$

Since $Y_{uv} = Y_{vu}$, the difference of their tangent components is zero. Thus

$$0 = \nabla_u \nabla_v Y - \nabla_v \nabla_u Y - h_v(Y)S(X_u) + h_u(Y)S(X_v),$$

which gives (2.1). □

3. Theorema Egregium

Theorem 3.1 (Gauss)**.** *The Gaussian curvature is determined by the first fundamental form and its first and second derivatives. Explicitly, with A as in Proposition 2.2,*

$$K = -\frac{FA_{11} + GA_{21}}{EG - F^2}.$$

Proof. By (2.1), $RX_u = fS(X_u) - eS(X_v)$. Taking the inner product with X_v and using the coefficients of II gives

$$\langle RX_u, X_v\rangle = f\langle S(X_u), X_v\rangle - e\langle S(X_v), X_v\rangle = f^2 - eg = -KD.$$

On the other hand, the coordinate column of X_u is $(1, 0)^t$. Proposition 2.2 therefore gives

$$RX_u = A_{11}X_u + A_{21}X_v, \qquad \langle RX_u, X_v\rangle = FA_{11} + GA_{21}.$$

Equating the two expressions and dividing by D proves the formula. By Proposition 1.3, the matrices Γ_1, Γ_2 are determined by E, F, G and their first derivatives. Thus A is determined by E, F, G and their derivatives of order at most two. The formula for K therefore shows that K is completely determined by the first fundamental form. □